Capstone Project Number -13

Here's a **structured capstone project outline** for *Traffic Congestion Forecasting using Sensor Data*. Each section has **10 interconnected, progressively complex questions**. The flow ensures learners use **Advanced Excel → SQL → Python → Data Science (non-ML) → Power BI** and finally build a **dashboard**.

Section 1: Advanced Excel (Data Preparation & Exploration)

1. Import raw sensor data (traffic counts, timestamps, vehicle types) into Excel.
2. Clean missing values and outliers using conditional formatting and formulas.
3. Create pivot tables to summarize traffic volume by hour and day.
4. Use advanced functions (INDEX-MATCH, SUMPRODUCT) to calculate congestion ratios.
5. Build a time-based trend chart for peak vs. off-peak traffic.
6. Apply data validation to categorize vehicle types (car, bus, truck).
7. Use array formulas to compute rolling averages of traffic flow.
8. Perform scenario analysis: What-if traffic increases by 20% during rush hours.
9. Create a dynamic dashboard in Excel with slicers for date and location.
10. Export cleaned and structured data for SQL integration.

Section 2: SQL (Data Management & Querying)

1. Design relational tables for sensor data (traffic volume, vehicle type, location).
2. Write queries to aggregate traffic counts by location and time.
3. Use window functions (ROW_NUMBER, LAG) to calculate congestion trends.
4. Create stored procedures to automate daily traffic summaries.
5. Write complex joins to combine sensor data with weather conditions.
6. Use CASE statements to classify congestion levels (low, medium, high).
7. Optimize queries with indexes for faster retrieval.
8. Generate hourly congestion reports using GROUP BY and HAVING.
9. Create views for traffic density by road segment.
10. Export query results into CSV for Python analysis.

Section 3: Python (Data Wrangling & Analysis)

1. Import SQL-exported data into Pandas Data Frames.

2. Clean and normalize data using Python functions.
3. Perform time-series decomposition (trend, seasonality, residuals).
4. Use correlation analysis to check impact of weather on congestion.
5. Write custom functions to calculate congestion severity scores.
6. Automate daily traffic summary reports with Python scripts.
7. Visualize traffic flow using Matplotlib/Seaborn.
8. Apply moving averages and exponential smoothing (non-ML forecasting).
9. Build modular Python scripts for ETL (Extract, Transform, Load).
10. Export processed datasets into CSV/Excel for Data Science stage.

Section 4: Data Science (Non-ML Analytics)

1. Perform descriptive statistics (mean, median, variance) on traffic volume.
2. Conduct hypothesis testing: Is congestion significantly higher on weekdays?
3. Use ANOVA to compare congestion across multiple road segments.
4. Apply regression-free forecasting (trend extrapolation, smoothing).
5. Perform clustering using rule-based thresholds (not ML).
6. Calculate congestion probability distributions.
7. Build decision matrices for traffic management interventions.
8. Conduct sensitivity analysis on traffic volume vs. weather conditions.
9. Create congestion indices combining multiple variables.
10. Prepare summarized datasets for Power BI visualization.

Section 5: Power BI (Visualization & Dashboard)

1. Import processed datasets from Excel/Python into Power BI.
2. Create interactive visuals for traffic volume by time and location.
3. Build heatmaps showing congestion hotspots.
4. Add slicers for filtering by date, vehicle type, and weather.
5. Create KPI cards for average congestion levels.
6. Build trend lines for hourly and daily traffic flow.
7. Integrate SQL queries directly into Power BI for live updates.
8. Create drill-through reports for specific road segments.

9. Design a congestion severity dashboard with traffic-light indicators.
10. Publish the final dashboard for stakeholders with role-based access.

Final Deliverable: Integrated Dashboard

Learners must combine all outputs into a **Power BI dashboard** that:

- Shows **real-time traffic congestion trends**.
- Allows filtering by **location, time, vehicle type, and weather**.
- Provides **decision-support KPIs** for traffic management authorities.
- Demonstrates the **end-to-end workflow**: Excel → SQL → Python → Data Science → Power BI.

This structure ensures **progressive complexity** and **interconnection** across tools, while avoiding machine learning.